And indeed, insertion, removal and search are guaranteed to be logarithmic on our balanced BST: much better than our initial linear time search, isn't it? A logarithmic running time grows amazingly slow, just think that for a million points, from a million distances to be computed, we would go down to just about 20!

Figure 8.6 shows how we would run binary search on the binary search tree in the previous figure, to find the nearest neighbor of a point whose distance from the origin (aka its x coordinate) is 75. If we had an exact match, the nearest neighbor would be the result of binary search. When we don't have an exact match, which is the most common case, then the nearest neighbor will always be either the node where binary search fails, or its parent.

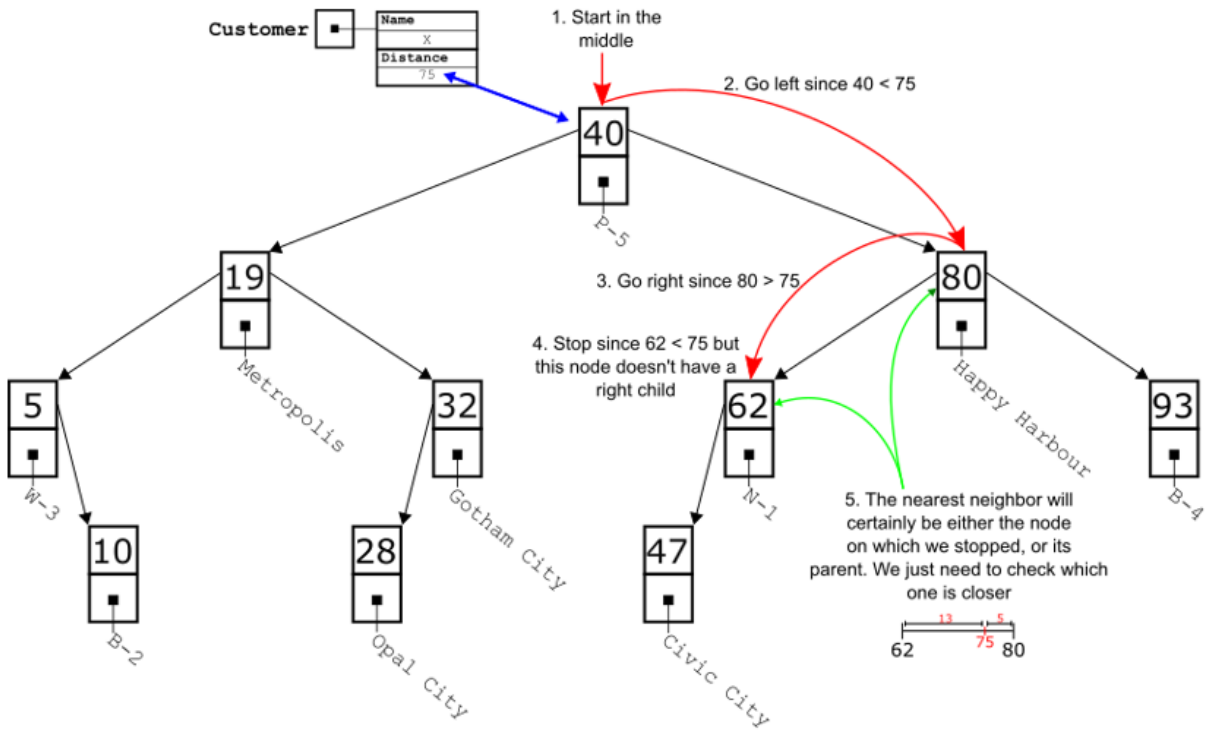


Figure 8.6 Using search on a binary search tree to find the nearest neighbor (in 1-D) of a target point.